



GROUPEMENT BERKINE

MARCH 10, 2026

2025 Annual GHG & CAP Emissions Report: HBNS and El Merk

OIL AND GAS PRODUCTION IN ALGERIA

Prepared by:

- Abdelhamid Hamidcha, Environmental Engineer / Reporting Team
- Wail Mouda, Environmental Engineer / Reporting Team

TABLE OF CONTENTS

TABLE OF CONTENTS	N°
Glossary	3
Executive Summary	4-6
• 2025 GHG Performance	4
• 2025 Methane Performance	5
• 2025 Flaring Performance	6
• 2025 Production, Operations, and Emissions Overview	6
• 2025 Improvements to Estimation Methodology	6
Key actions	7
Yearly Production Trend	8
Yearly Total GHG Emissions By Scope	9
Yearly Total GHG Emissions By module	10-12
• HBNS	10
• ELM	11
• GB	12
GB Total GHG Emissions Progress Towards Target	13
Yearly total CH4 Emissions	14
CPF Flaring Emissions	15-18
• HBNS	15
• ELM	16
• GB	17
• Flaring Intensity	18
Carbon Intensities	19
Equity Share Emissions	20-21
• Total GHG	20
• Total CH4	21
CAP Emissions by source module	22
Appendices: Data Tables	23-41

APPENDICES	N°
Production – 2021 to 2025	24-25
• HBNS	24
• ELM	25
GHG Emissions by Module – 2021 to 2025	26-28
• HBNS	26
• ELM	27
• GB	28
CH4 Emissions by Module – 2021 to 2025	29-31
• HBNS	29
• ELM	30
• GB	31
2025 Yearly CPF Flaring Breakdown	32
2021 to 2025 Total Production, GHG, Methane, Flare Volumes & Intensities –	33-35
• HBNS	33
• ELM	34
• GB	35
Equity Share Emissions	36-37
• GHG	36
• CH4	37
CAP Emissions by Source Module 2021 to 2025	38-41
• HBNS	38-39
• ELM	40-41

GLOSSARY

TERM	DEFINITION	TERM	DEFINITION
API	American Petroleum Institute	JV PARTNER	Joint Venture Partner
AR4	Fourth Assessment Report (Intergovernmental Panel on Climate Change)	LDAR	Leak Detection and Repair
AR5	Fifth Assessment Report (Intergovernmental Panel on Climate Change)	LPG	Liquefied Petroleum Gas
BOE	Barrel of Oil Equivalent	NGL	Natural Gas Liquids
CAP	Criteria Air Pollutant	NGSI	Natural Gas Sustainability Initiative
CO2e	Carbon Dioxide Equivalent	OGCI	Oil and Gas Climate Initiative
CH4	Methane	OGDC	Oil & Gas Decarbonization Charter
CPF	Central Processing Facility	OGI	Optical Gas Imaging
DRE	Destruction and Removal Efficiency	OGMP2.0	Oil and Gas Methane Partnership 2.0
ELM	El Merk	PM	Particulate Matter
FGS	Field Gathering Station	PQLA	Process Quick Look Assessment
GHG	Greenhouse Gase	VISR	Video Imaging Spectral Radiometer
GWP	Global Warming Potential	VOC	Volatile Organic Compounds
HBNS	Hassi Berkine	ZRF	Zero Routine Flaring

EXECUTIVE SUMMARY

2025 GHG PERFORMANCE

■ 2025 Progress Towards GHG Emissions Goals:

- Total GHG emissions (CO₂e) – achieved total reduction of 15.9% relative to the 2.27 million-tonnes baseline (2021–2023 average).
- Net GHG emissions – achieved reduction of 3.5% in 2025, improving on 12.4% achieved in 2024.
- Future initiatives 2026 to 2030 - plan for additional 0.32 million-tonnes reductions in total GHG emissions, to achieve continue progress towards the 2030 target of 1.59 million-tonnes annually (30% reduction by 2030 target).

■ 2025 Total GHG Emissions Decreased 4.0% Compared to 2024:

- HBNS CPF flaring reduction milestones: achieved lowest annual flaring volume ever recorded, and reduced below 1.0% threshold.
- Application of measured DRE results on all CPF flares.

■ 2025 GHG Intensity based on Total Production Decreased 1.5 % Compared to 2024:

- Carbon intensity per site :
 - HBNS - 11.2 kg CO₂e/BOE
 - ELM - 18.5 kg CO₂e/BOE

■ 2025 GHG Intensity based on Saleable Products Decreased 0.9% Compared to 2024:

- Carbon intensity per site:
 - HBNS - 30.8 kg CO₂e/BOE
 - ELM - 36.1 kg CO₂e/BOE

EXECUTIVE SUMMARY

2025 METHANE PERFORMANCE

- Completed 2nd annual LDAR campaign, with incorporation of high-flow sampler measurements for > 99% of detected fugitive leaks.
- Integrated LDAR campaign high-flow sampler measurements into fugitive emissions source reporting.
- Application of measured flare DRE measurements for both CPF flare networks.
- **Methane Emissions Decreased 40% Compared to 2024.**
- **Methane Intensity:**
 - Site specific methane intensities, calculated following NGSi method :
 - HBNS 0.05%
 - ELM 0.03%
 - **GB Methane Intensity based on Total Production decreased 35% from 2024 to 2025:**
 - **GB Methane Intensity based on Saleable Products decreased 35% from 2024 to 2025.**
 - **GB Methane Intensity based on NGSi Method (wt.%) decreased 42% from 2024 to 2025.**

NOTE 1: Transition to measurement based methodology for fugitives leaks in 2025 (versus component type emission factors - 2024) resulted in 189% increase from 2024. By comparison, this highest-accuracy method still represents a 92.3% reduction relative to the 2021–2023 baseline.

NOTE 2: Ongoing development and improvements in the GB LDAR program are anticipated to result in lower future emissions as repair is prioritized and survey frequency increases.

NOTE 3 : In 2025, measured destruction removal efficiency (DRE) results were applied to all CPF flares. This replaces prior years' use of the standard 98% default, with DRE now determined through annual field measurements using VISR camera.

EXECUTIVE SUMMARY

2025 FLARING PERFORMANCE

HBNS:

- CPF flaring volume decreased by 24.91% in 2025, compared to 2024.
- Routine flaring remains the largest component of CPF gas flared: the ratio increased to 56.2% in 2025, compared to 41.6% in 2024.
- HBNS Total Flaring intensity decreased to 0.91% in 2025, from 1.17% in 2024.

ELM:

- CPF flaring volume increased by 9.28% in 2025 compared to 2024 due to the 3 year planned full CPF shutdown.
- Routine flaring remains the largest component of CPF gas flared: the ratio decreased to 68.6% in 2025, compared to 78.2% in 2024.
- ELM Total Flaring intensity increased to 1.63% in 2025, from 1.45% in 2024.

2025 PRODUCTION, OPERATIONS, AND EMISSIONS OVERVIEW

- Total BOE production decreased by 1.4% in 2025 compared to 2024, and overall emissions reflect a corresponding decrease.
 - HBNS production increased 9.2 %.
 - ELM production decreased 8.2%.

2025 IMPROVEMENTS IN ESTIMATION METHODOLOGY

- Measured destruction removal efficiency (DRE) between 99.74% and 99.89% was applied to all CPF flares. This replaces prior years' use of the standard 98% default, with DRE now determined through annual field measurements using VISR camera.

KEY ACTIONS

Commitment to Zero Routine Flaring (ZRF):

- Completed concept-selection phase for Flare Gas Recovery Units and started Front-End Engineering Design for both HBNS and ELM CPF flare networks.

Commitment to OGMP 2.0

- Completed level 4 LDAR campaign across both HBNS and ELM fields (included CPF's, wells, FGS's). OGI camera used to detect leaks followed by high-flow sampling to quantify accessible leaks.
- Completed Level 4 Flare DRE measurements utilizing VISR camera technology for all operated flares at both HBNS & ELM CPF's.

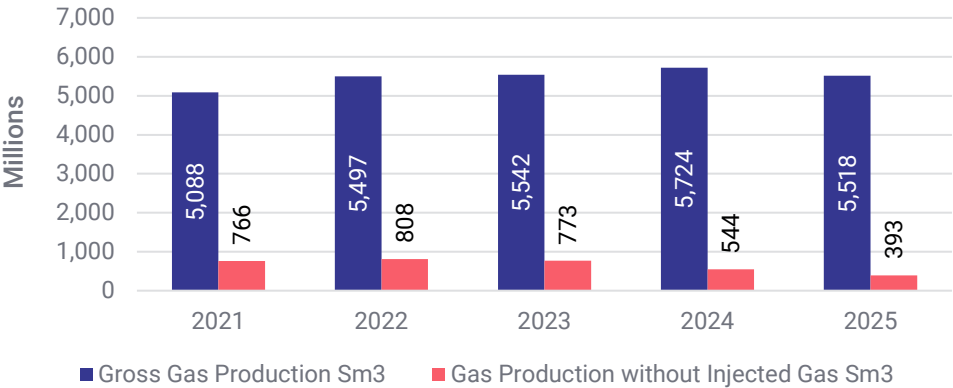
GHG Reduction Commitments:

- Front-End Engineering Design in progress for ~35.6 MWp photovoltaic plant at HBNS CPF.
- Front-End Engineering Design in progress for ~20 MWp photovoltaic plant at ELM CPF.
- Completed feasibility study for Waste Heat Recovery Unit installation at HBNS CPF .
- Application of flaring policy for both CPF's.
- PQLA (Process Quick Look Assessment) completed at HBNS CPF with 44 potential actions identified to reduce emissions or increase accuracy of reported GHG emissions.
- Reduction in reported methane and CO2e from CPF flares as a result of DRE measurement campaign.

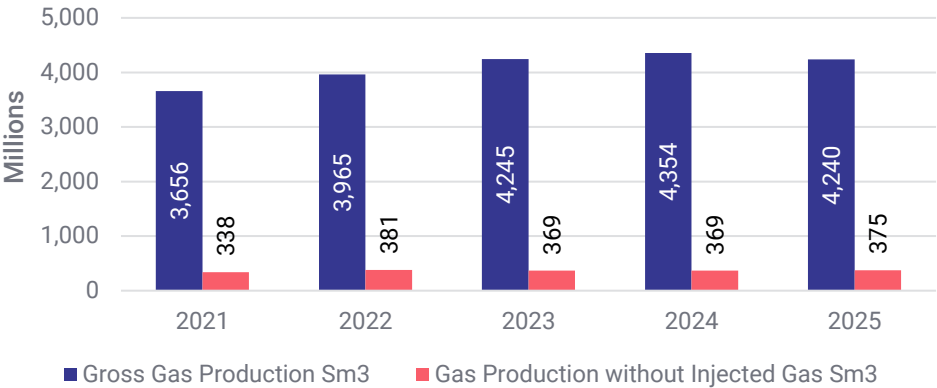
YEARLY PRODUCTION TREND



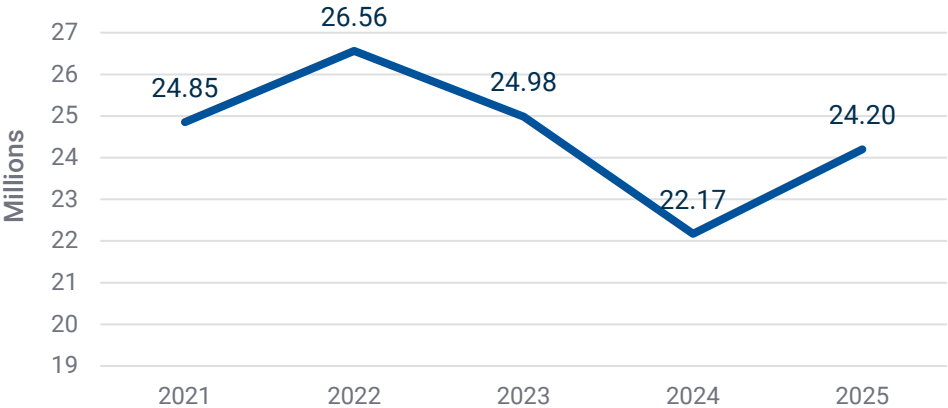
HBNS Gas Production (Sm3)



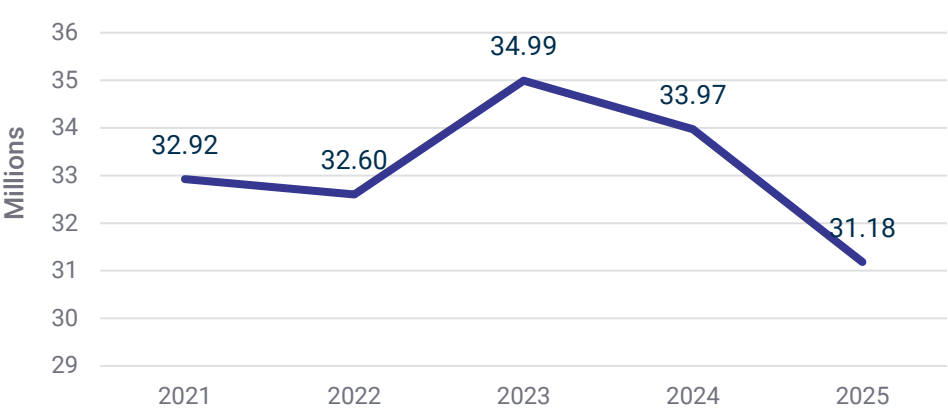
ELM GAS Production (Sm3)



HBNS Liquid Production (BOE)



ELM Liquid Production (BOE)

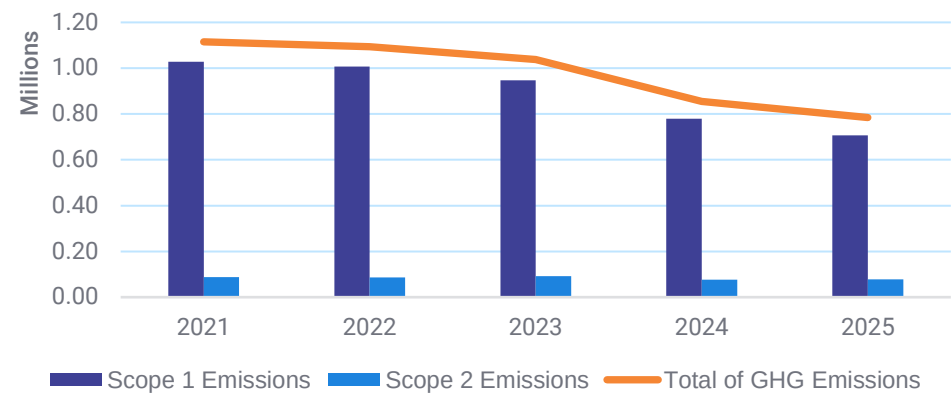


YEARLY TOTAL GHG EMISSIONS BY SCOPE

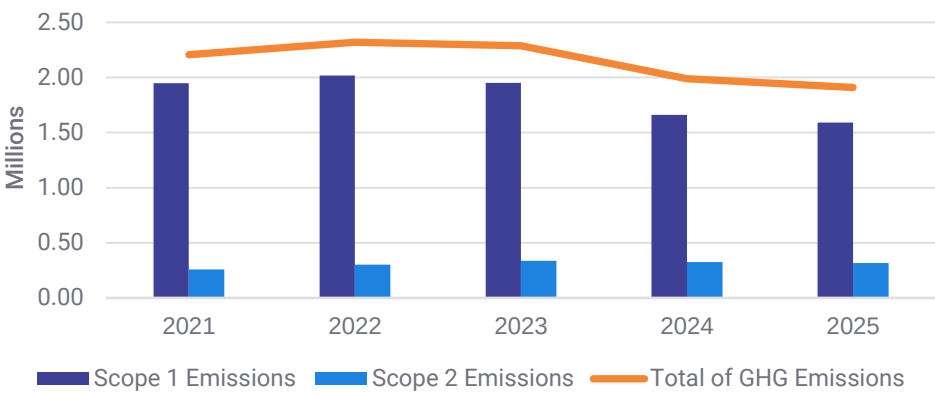
Total CO2e Emissions (Tonnes/Year)



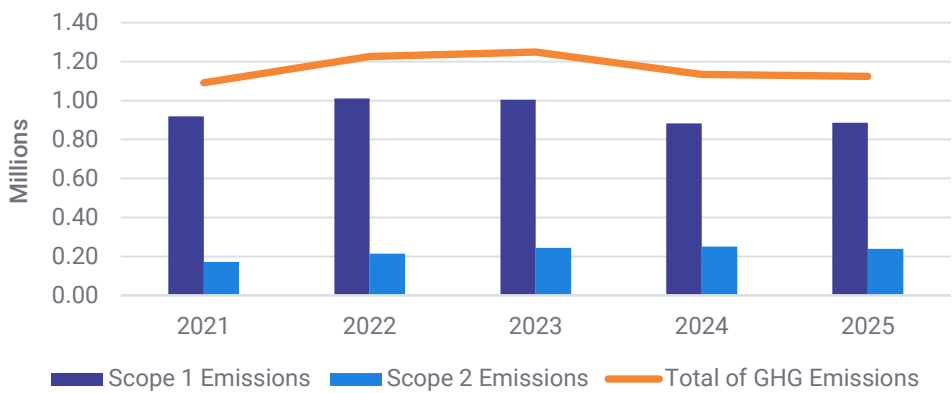
HBNS YEARLY TREND



GB TOTAL YEARLY TREND



ELM YEARLY TREND



KEY OBSERVATIONS

- GB Total GHG Emissions showed an overall decline, decreasing from 2021 to 2025, primarily related to reductions at HBNS.

*Calculations use AR5 GWP for 2024 and 2025 and AR4 GWP for 2021–2023.

**Scope 2 emissions are calculated using MWh data and the PK0 turbine efficiency factor. The switch from the national grid to PK0 efficiency was made in 2023, with all reports from 2021 to 2023 updated accordingly.

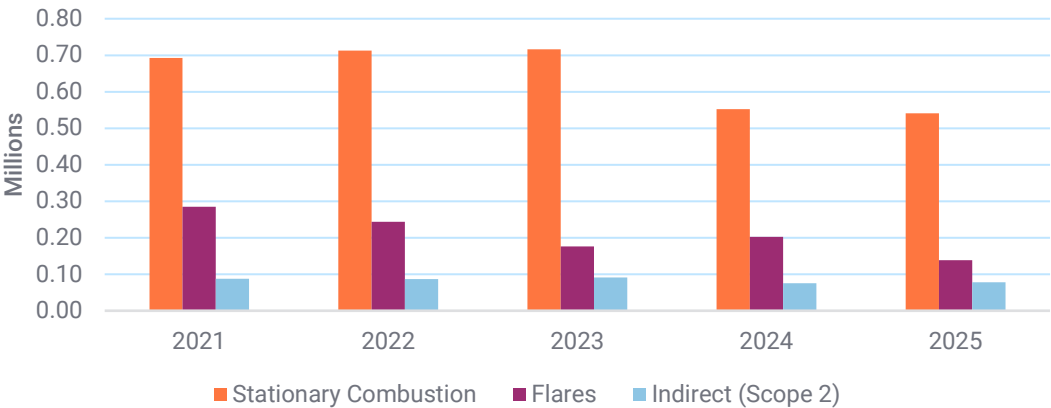


HBNS YEARLY TOTAL GHG EMISSIONS BY MODULE

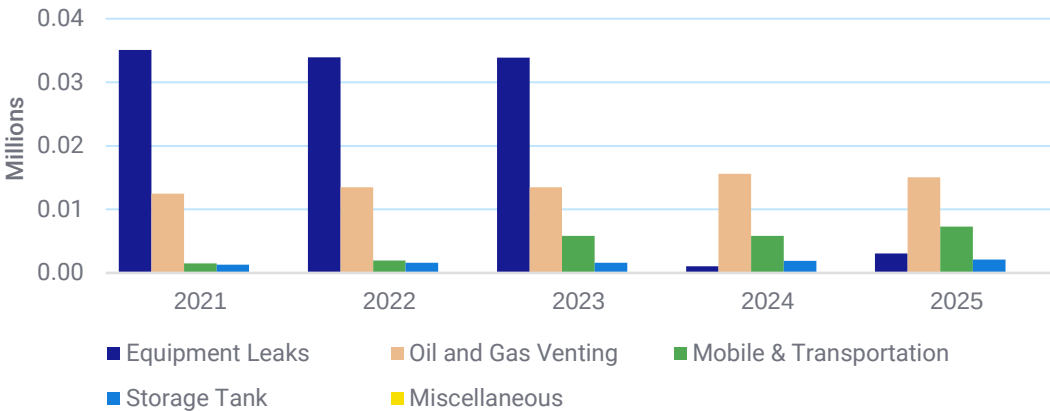
Total CO2e Emissions (Tonnes/Year)



HBNS YEARLY GHG TREND BY MODULE



HBNS YEARLY GHG TREND BY MODULE



KEY OBSERVATIONS

- Stationary Combustion – 2 % decrease in 2025 compared to 2024.
- Equipment Leaks – 136% increase in 2025 compared to 2024 [Refer to NOTE 1 in Executive Summary].
- Indirect Emissions (Scope 2) – 3% increase in 2025 compared to 2024.
- Flare Emissions decrease – 32 % in 2025 compared to 2024 [Refer to NOTE 3 in Executive Summary].

*Calculations use AR5 GWP for 2024-2025 and AR4 GWP for 2021 to 2023.

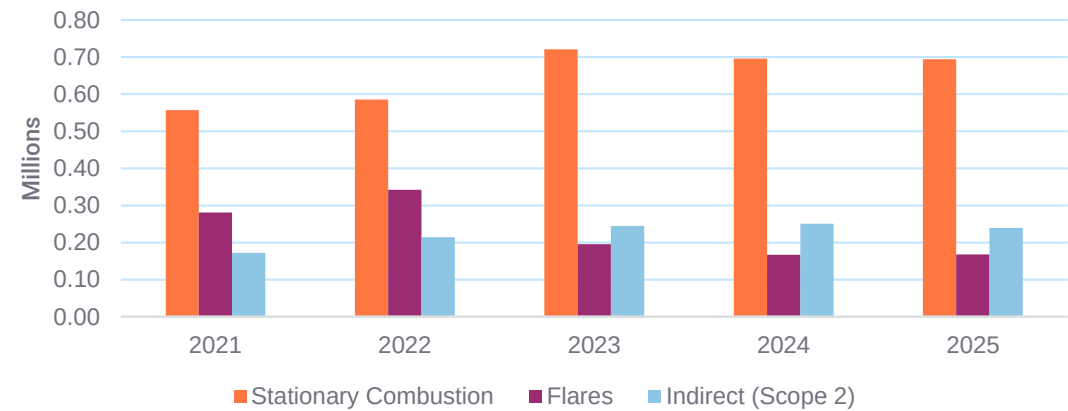


ELM YEARLY TOTAL GHG EMISSIONS BY MODULE

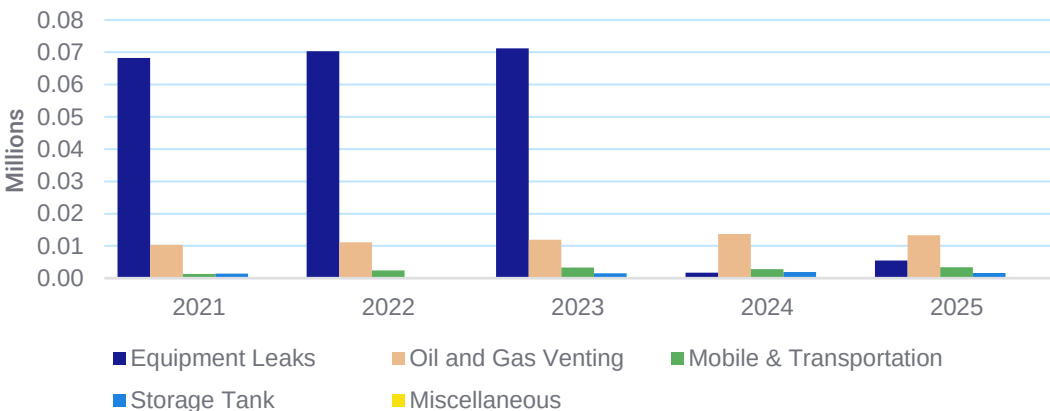
Total CO2e Emissions (Tonnes/Year)



ELM YEARLY GHG TREND BY MODULE



ELM YEARLY GHG TREND BY MODULE



KEY OBSERVATIONS

- Equipment Leaks – 221% increase in 2025 compared to 2024 [Refer to NOTE 1 in Executive Summary].
- Flaring Emissions – 1% increase in 2025 compared to 2024 (Due to 3 year planned full CPF shutdown).
- Indirect Emissions (Scope 2) – 4% decrease in 2025 compared to 2024.

*Calculations use AR5 GWP for 2024-2025 and AR4 GWP for 2021 to 2023.

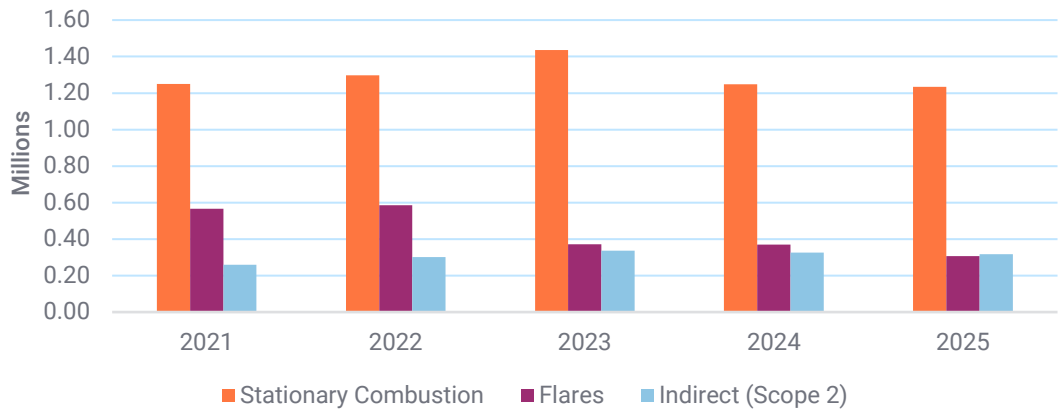


GB TOTAL YEARLY GHG EMISSIONS BY MODULE

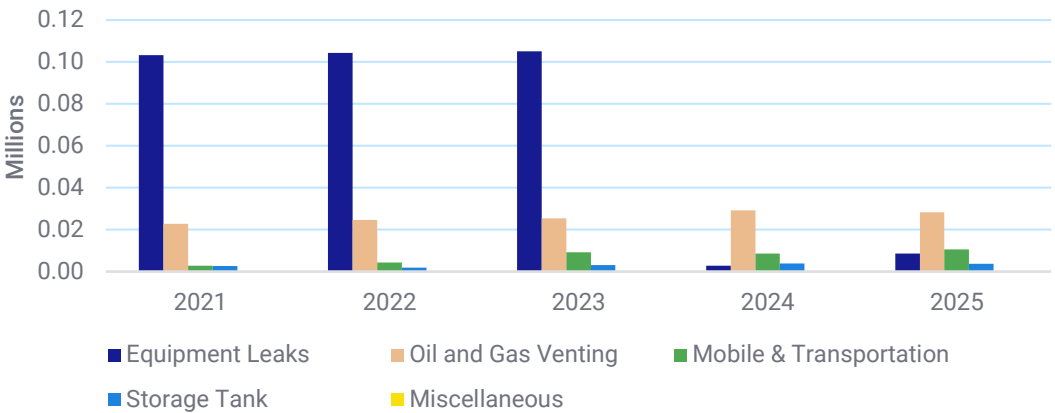
Total CO2e Emissions (Tonnes/Year)



GB TOTAL YEARLY GHG TREND BY MODULE



GB TOTAL YEARLY GHG TREND BY MODULE



KEY OBSERVATIONS

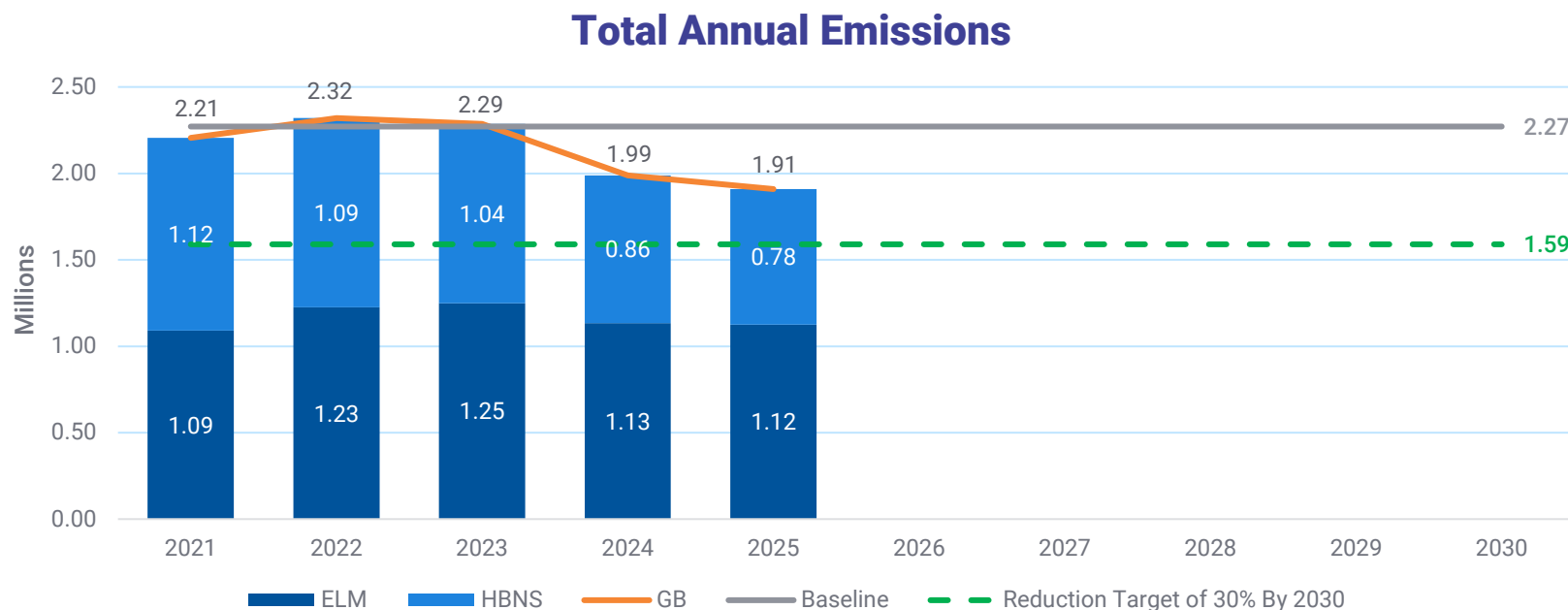
- Flaring emissions – 17% decrease in 2025, compared to 2024.
- Equipment Leaks – 209% increase [Refer to NOTE 1 in Executive Summary].
- Indirect Emissions (Scope 2) – 3% decrease in 2025 compared to 2024.

*Calculations use AR5 GWP for 2024-2025 and AR4 GWP for 2021 to 2023.



GB TOTAL GHG EMISSIONS PROGRESS TOWARDS TARGET

Total CO2e Emissions (Tonnes/Year)



KEY OBSERVATIONS

- In 2025 GB reduced Total GHG Emission 16% below the set baseline of 2.27 million tonnes (average of 2021 to 2023).
- Corresponding 2025 Total GHG Emissions of 1.91 million tonnes is 0.32 million tonnes above the 2030 reduction target of 1.59 million tonnes.
- 2026 to 2030 focus is further reductions of at least 0.32 million tonnes in Total GHG Emissions, to achieve the 30% reduction target of 1.59 million tonnes (annual Total GHG Emissions).

*Baseline is the average of 2021, 2022, and 2023 GHG emissions.

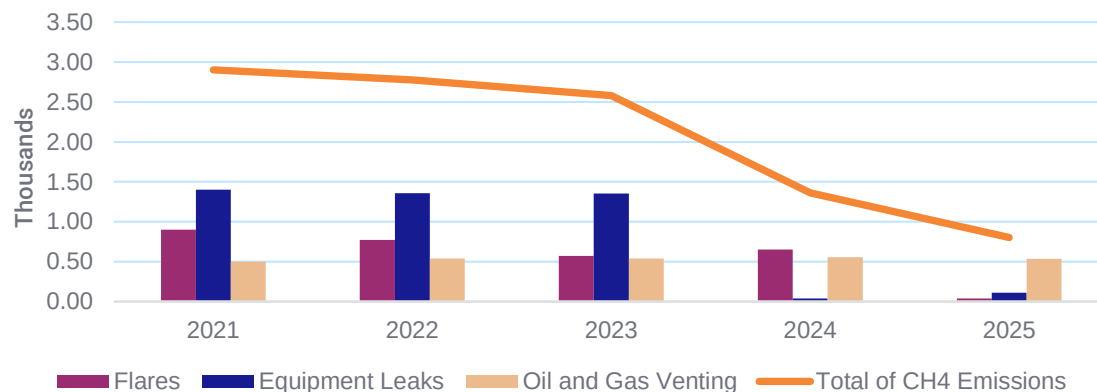


YEARLY TOTAL CH4 EMISSIONS

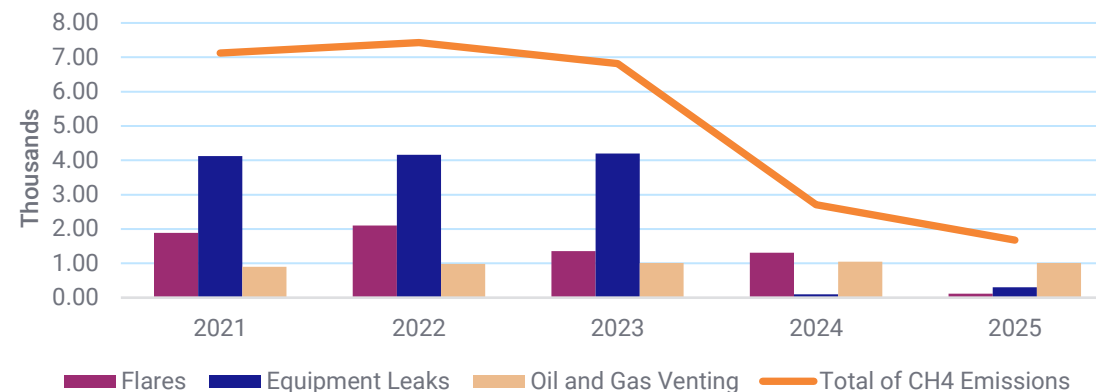
Total CH4 Emissions (Thousand Tonnes/Year)



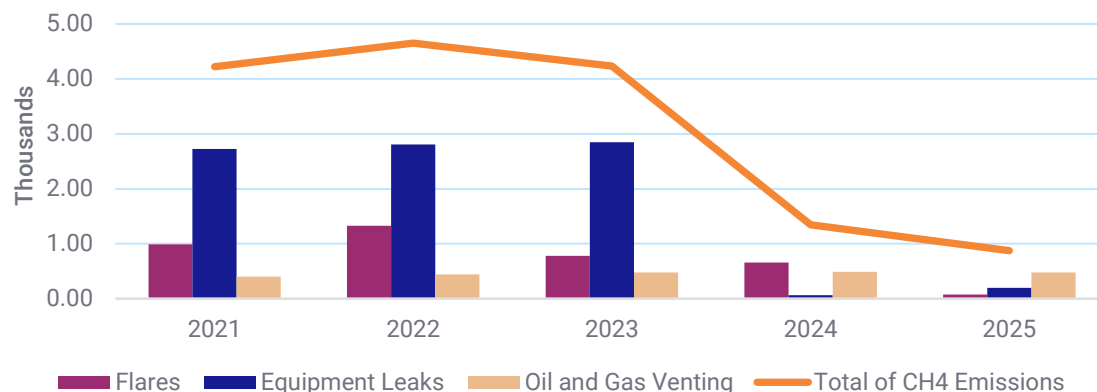
HBNS YEARLY TREND



GB TOTAL YEARLY TREND



ELM YEARLY TREND



KEY OBSERVATIONS

- GB CH4 Emissions reduced by 39% from 2024 to 2025, related to:
 - Flares - 91% decrease
- Both HBNS and ELM contributed to the overall GB decrease:
 - HBNS - 42% decrease
 - ELM - 35% decrease

*Calculations use AR5 GWP for 2024 and AR4 GWP for 2021–2023.

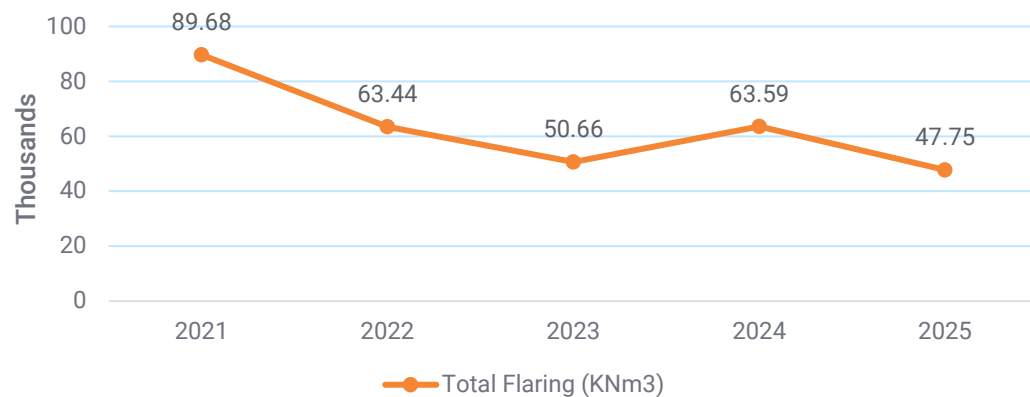


HBNS CPF FLARING BREAKDOWN

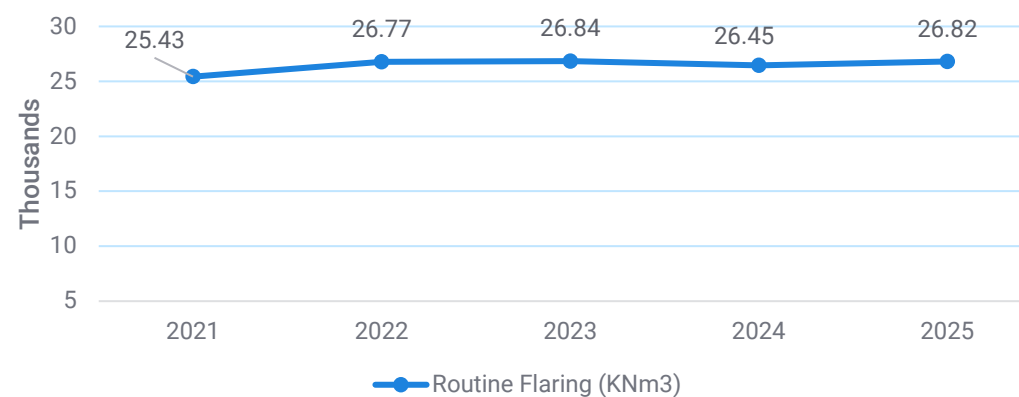
Flaring in KNm3



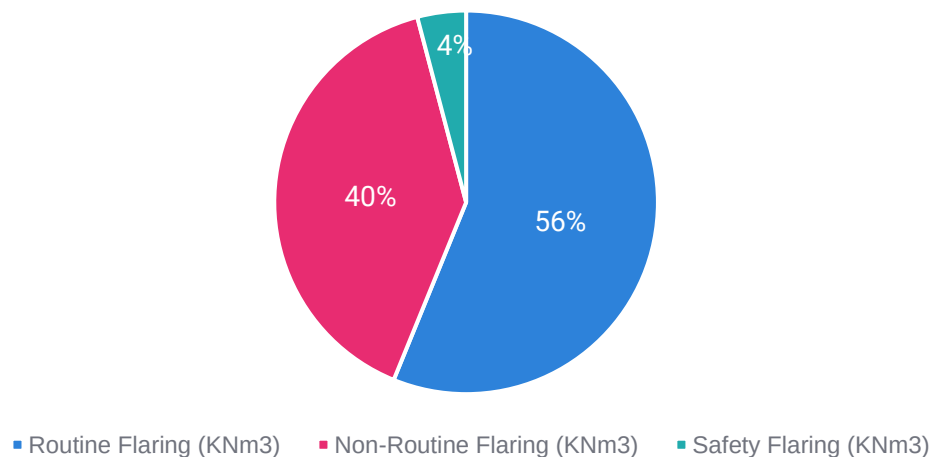
HBNS Total Flaring



HBNS Routine Flaring



HBNS 2025



KEY OBSERVATIONS

- Routine flaring - primary contributor in 2025 .
- Routine flaring increased by 1.4% in 2025 compared to 2024 volumes.

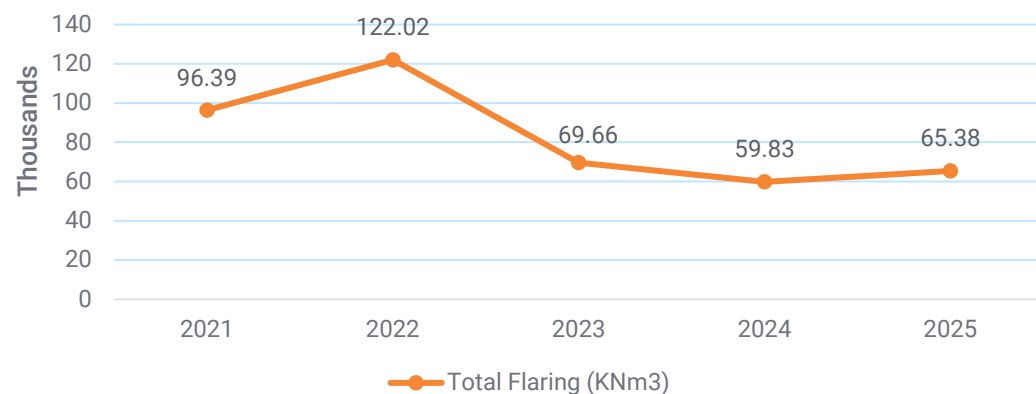


ELM CPF FLARING BREAKDOWN

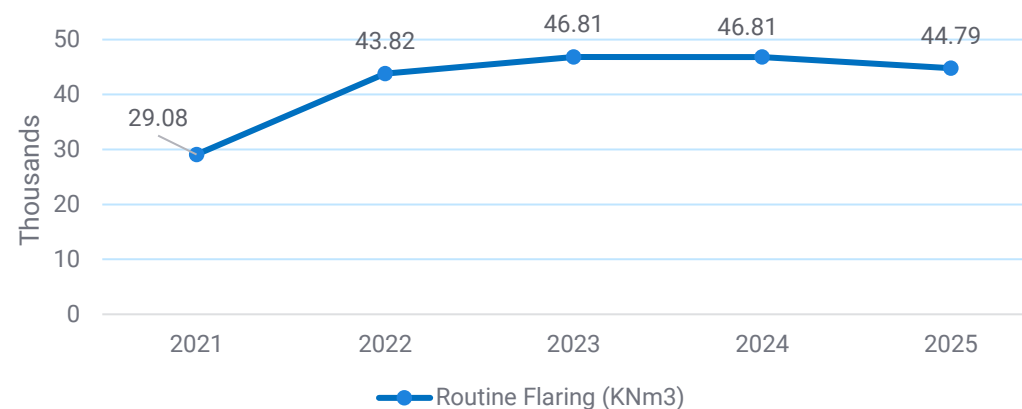
Flaring in KNm3



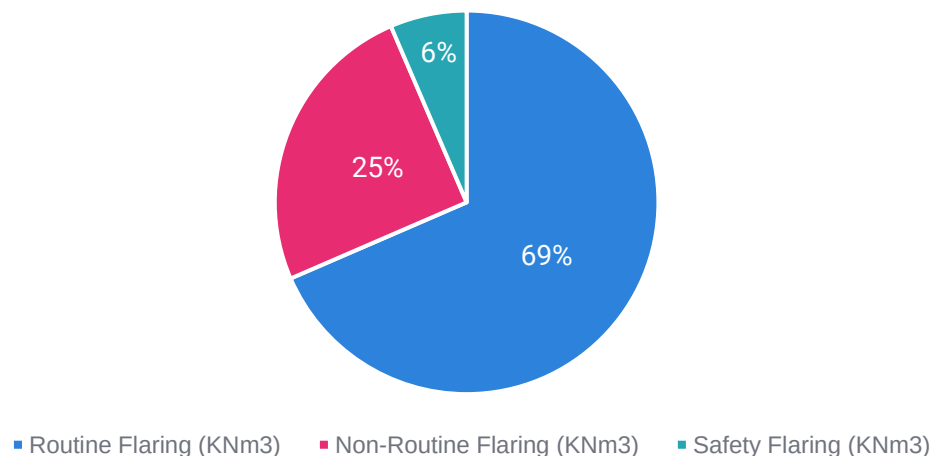
ELM Total Flaring



ELM Routine Flaring



ELM 2025



KEY OBSERVATIONS

- Routine flaring - primary contributor in 2025 .
- Routine flaring decreased by 4.3% in 2025 compared to 2024 volumes.

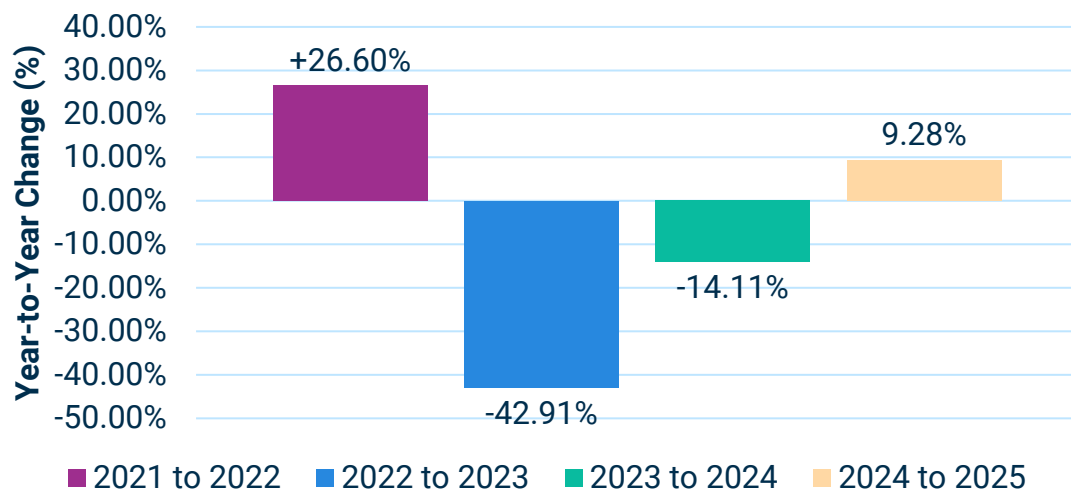
*From 2022 onwards, the routine, non-routine, and safety flaring classification for ELM was modified and updated using recorded events and volumes.



TOTAL CPF FLARING YEARLY TREND



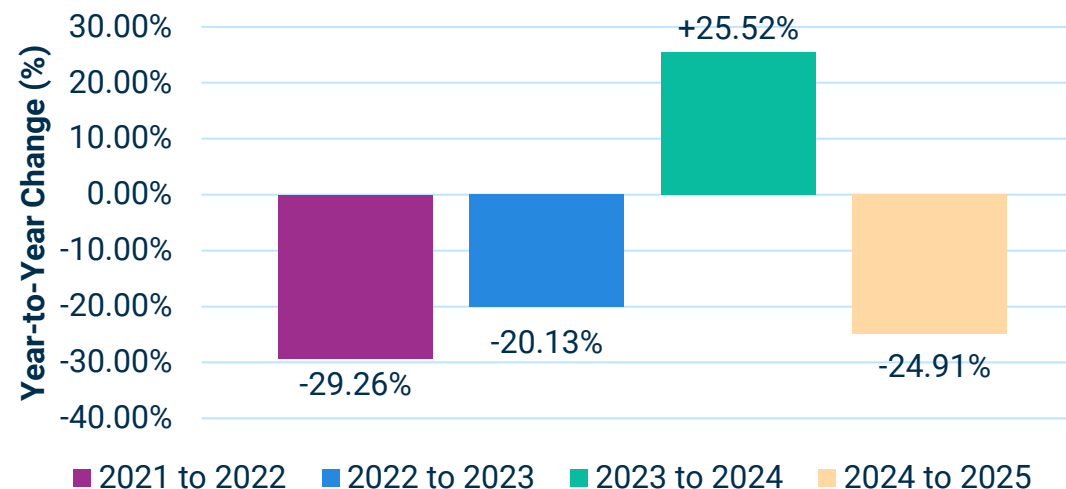
ELM Year-to-Year Percentage Change in Total Flaring volume



KEY OBSERVATIONS

- 26.60% increase from 2021 to 2022 related to 3 year planned CPF shutdown.
- 42.9% decrease from 2022 to 2023.
- Continued 14.11% decrease from 2023 to 2024 (application of flaring policies).
- 9.28% increase from 2024 to 2025 related to 3 year planned CPF shutdown (Large improvement in shutdown related flaring performance compared to 2022 shutdown).

HBNS Year-to-Year Percentage Change in Total Flaring volume



KEY OBSERVATIONS

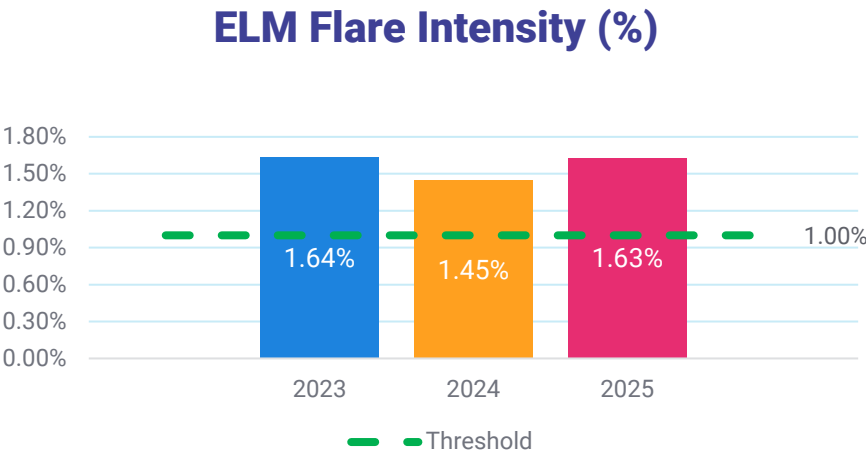
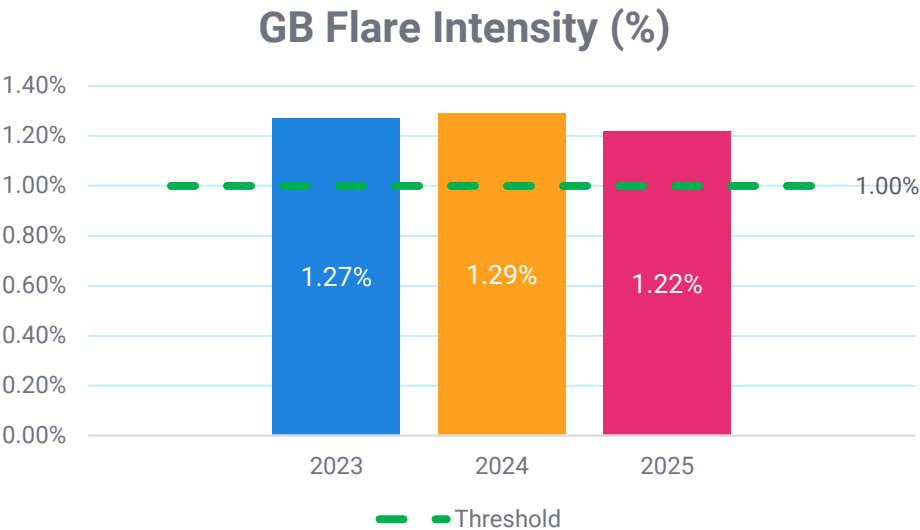
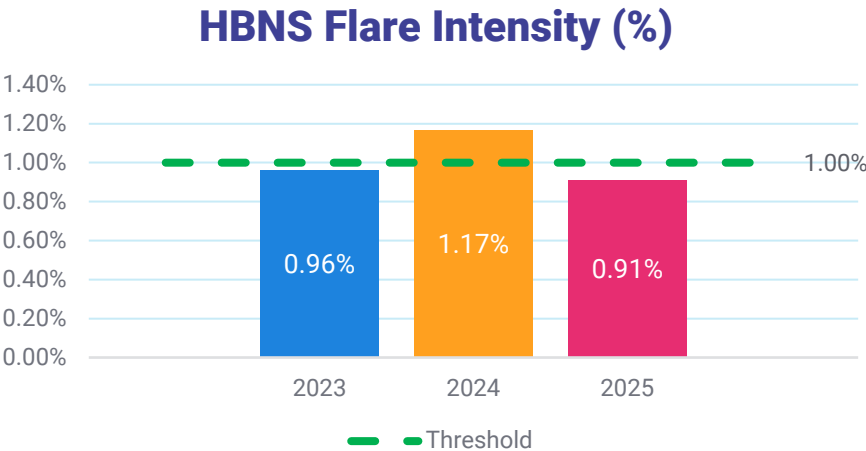
- 29.26% decrease from 2021 to 2022 (2021 3 year planned CPF shutdown).
- Further 20.13% decrease from 2022 to 2023 (application of flaring policies).
- 25.52% increase from 2023 to 2024 related to 3 year planned CPF shutdown.
- 24.91% decrease from 2024 to 2025 (lowest flaring in CPF history resulting from operational excellence and flare network maintenance).

*Total flare includes Routine, Non-routine and Safety flaring volumes.



YEARLY CPF FLARING INTENSITY

Intensity (%)



KEY OBSERVATIONS

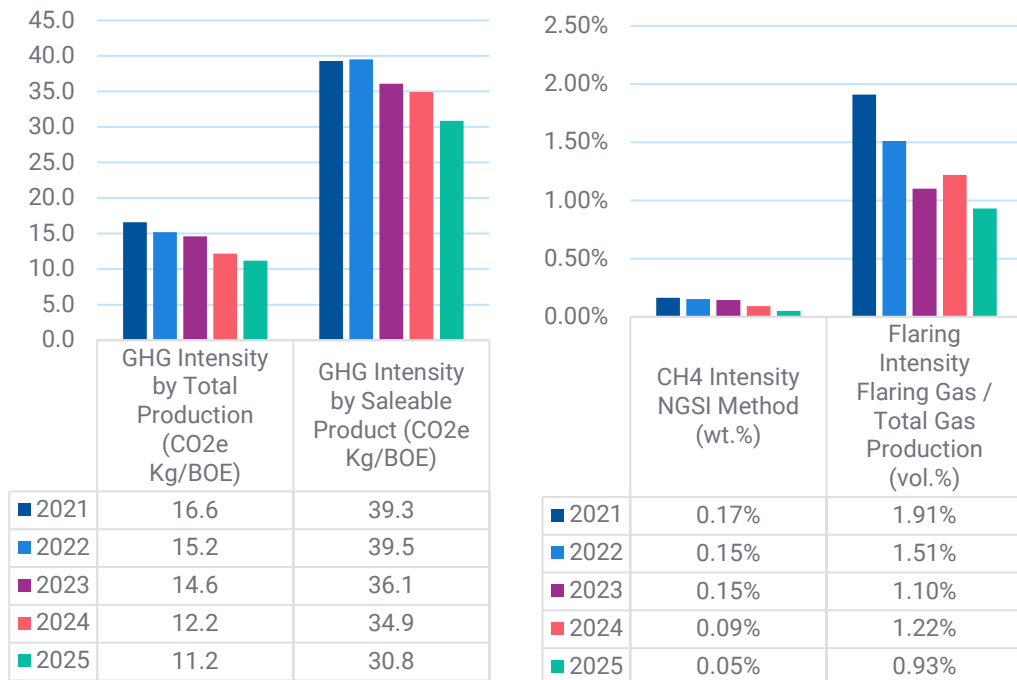
- HBNS Flaring Intensity decreased from 1.17% to 0.91%, Under the 1.00% Target .
- ELM Flaring Intensity increased from 1.45% to 1.63%, influenced primarily by ELM CPF shutdown.

*Total Flaring Intensity refers to percentage of total flared volume proportional to total gas produced, as defined in Article 9 of Executive Decree No. 21-330.

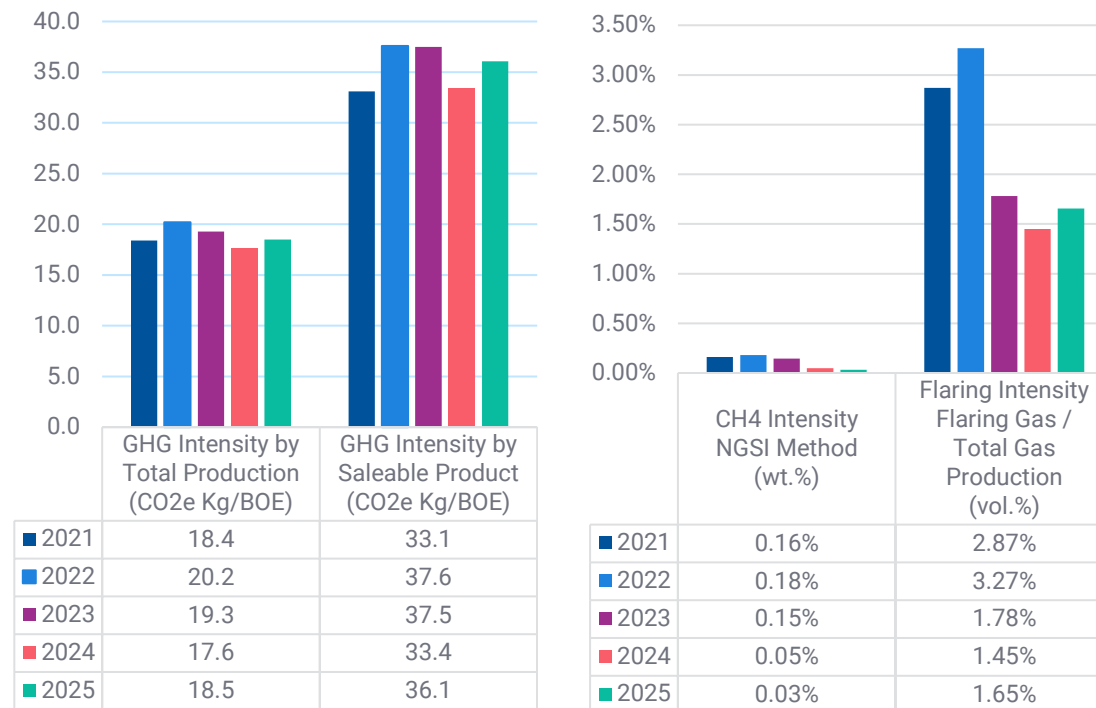


CARBON INTENSITIES

Summary of HBNS Intensities



Summary of ELM Intensities



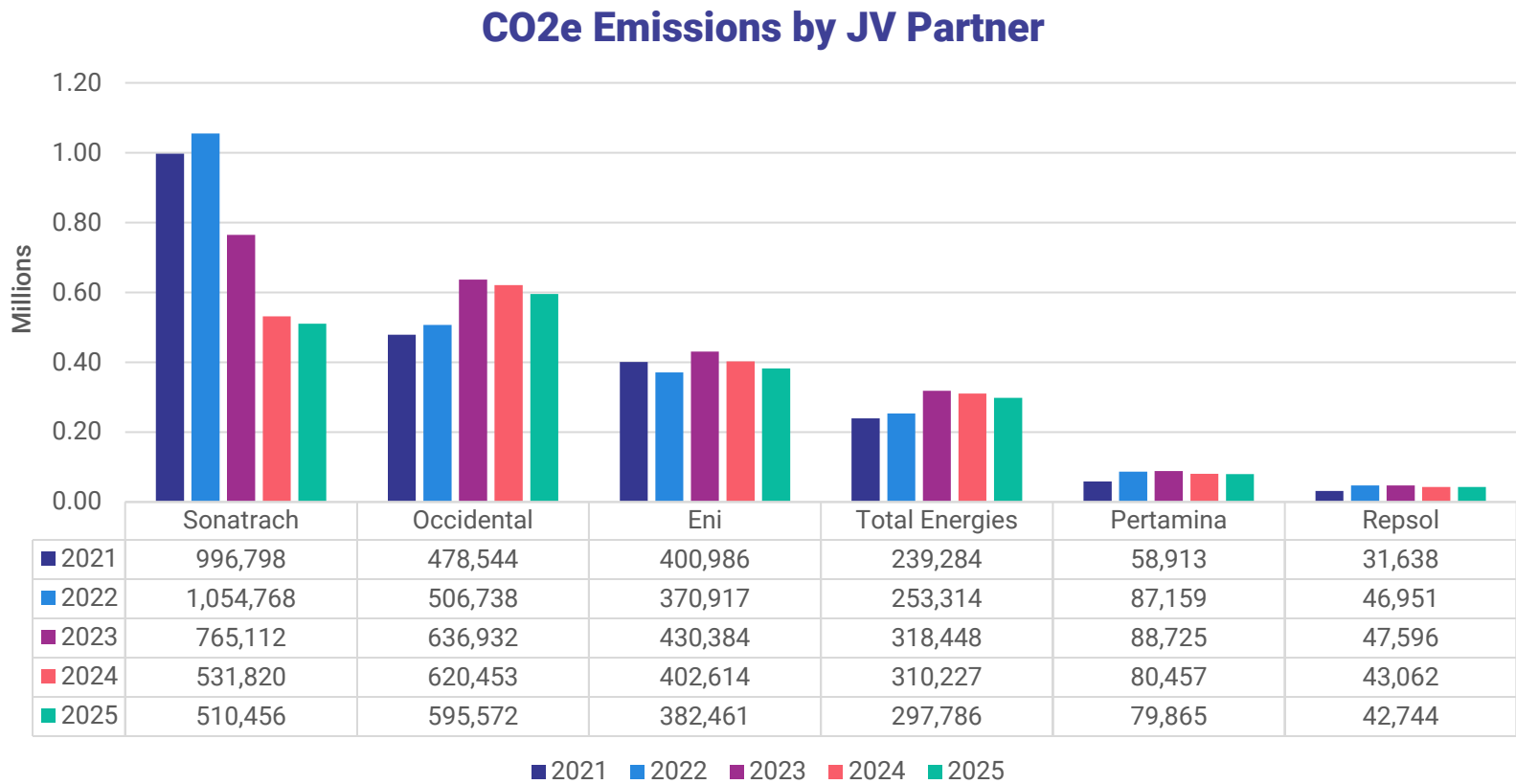
KEY OBSERVATIONS

- GHG Intensity by Total Production calculation method has shown a steady decrease at HBNS, while ELM fluctuates .
- GHG Intensity by Saleable Product shows reductions at HBNS for 2025. While ELM increased.
- Carbon intensity based on Saleable Product for HBNS and ELM are 30.8 and 36.1 Kg CO2e/BOE, respectively. These results are approximately twice above the OGCI recommended level of 17.0 Kg CO2e/BOE for 2025.



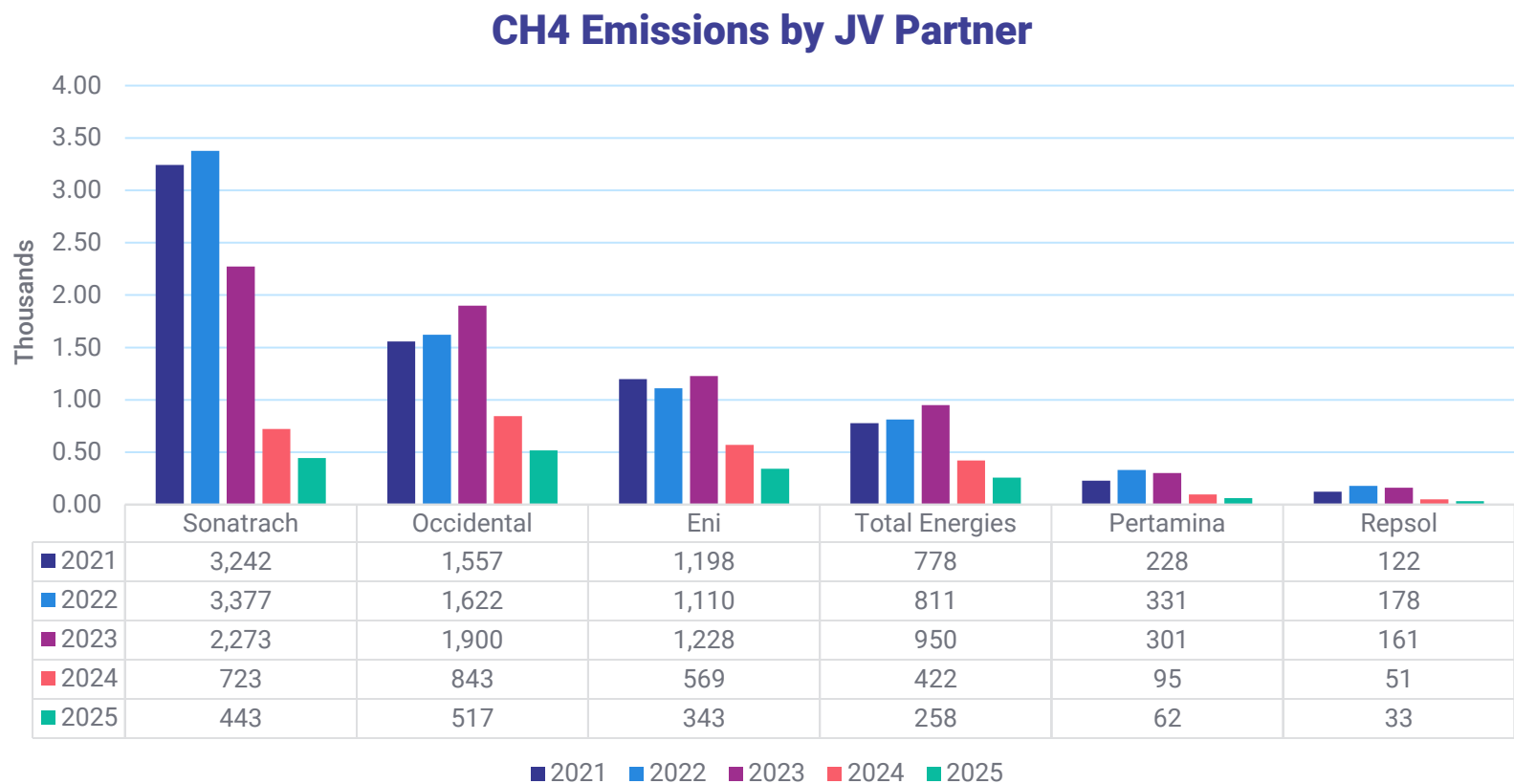
TOTAL GHG EMISSIONS BY JV PARTNER EQUITY SHARE

Total CO2e Emissions (Tonnes/Year)



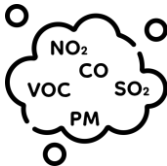
TOTAL CH4 EMISSIONS BY JV PARTNER EQUITY SHARE

Total CH4 Emissions (Tonnes/Year)

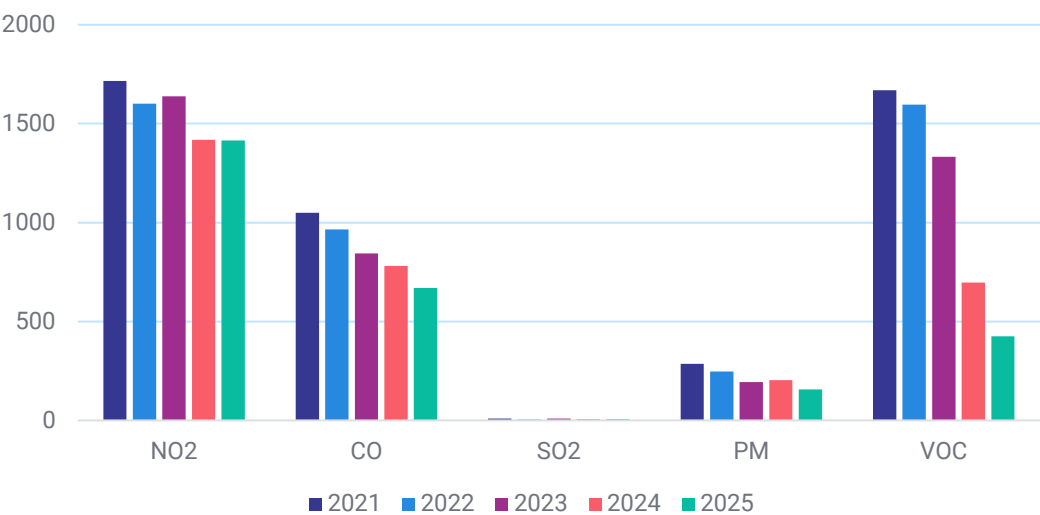


CAP EMISSIONS BY SOURCE MODULE

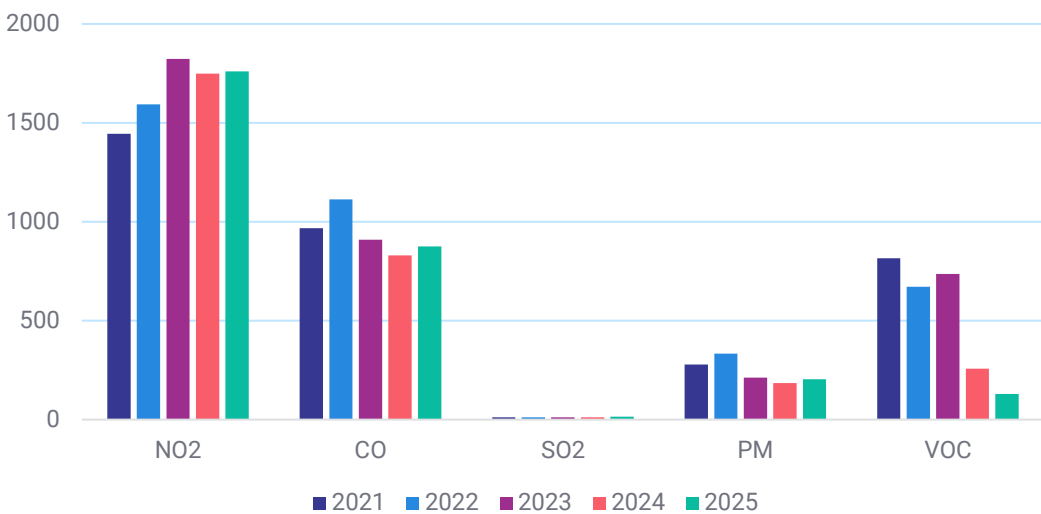
CAP Emissions (Tonnes/Year)



HBNS



ELM



2025 CAP Compliance

Site	HBNS			ELM		
CAP	Tonnes	Mg/Nm3	Limit Mg/Nm3	Tonnes	Mg/Nm3	Limit Mg/Nm3
NO2	1,414.70	788.00	200	1,760.78	980.77	200
CO	668.58	7.50	150	875.66	9.84	150
SO2	6.35	0.00	800	14.08	0.00	800
PM	156.91	0.00	30	204.26	0.00	30
VOC	406.37	11.13	150	130.45	3.41	150

KEY OBSERVATIONS

- Reduction in VOC and CO emissions for both HBNS and ELM in 2025.
- PM emissions fluctuated but showed a notable decrease in 2025.
- SO2 emissions remained relatively stable, with minor variations for both HBNS and ELM.
- The 2025 CAP compliance results show that the NO₂ limit has been exceeded. Additional sampling has been conducted on-site to confirm the results.

*Limits per the executive decree 06-138



APPENDICES

DATA TABLES



HBNS PRODUCTION – 2021 TO 2025



Products	Unit	2021	2022	2023	2024	2025
Gross Gas Production	MMSm3	5,088.03	5,497.26	5,541.74	5,723.88	5,517.62
	MMBOE	42.26	45.62	46.16	47.68	45.96
Gas Production without Injected Gas	MMSm3	765.51	808.06	772.85	544.06	392.69
	MMBOE	6.36	6.71	6.44	4.53	3.27
Liquid Production – Crude, Condensates, LPG	MMBOE	24.85	26.56	24.98	22.17	24.20
Total Production [1]	MMBOE	67.11	72.18	71.15	69.85	70.16
Total Production without Injected Gas [2]	MMBOE	31.21	33.27	31.42	26.70	27.47
Total Saleable Production [3]	MMBOE	28.36	30.45	28.77	24.50	25.44

[1] Total production includes crude oil, condensates, NGL, LPG, and total produced gas used for reinjection, artificial lifting, fuel gas, and flares.

[2] Total gas production without gas used for reinjection and artificial lifts.

[3] Saleable Production includes fuel gas exported to PK0, crude oil, condensates, NGL, and LPG.



ELM PRODUCTION – 2021 TO 2025



Products	Unit	2021	2022	2023	2024	2025
Gross Gas Production	MMSm3	3,655.66	3,964.74	4,244.53	4,353.68	4,239.89
	MMBOE	26.27	28.10	29.75	30.46	29.58
Gas Production without Injected Gas	MMSm3	338.35	380.95	369.26	369.25	374.81
	MMBOE	2.43	2.71	2.59	2.58	2.62
Liquid Production – Crude, Condensates, LPG	MMBOE	32.92	32.60	34.99	33.97	31.18
Total Production [1]	MMBOE	59.19	60.71	64.74	64.43	60.77
Total Production without Injected Gas [2]	MMBOE	35.35	35.32	37.58	36.56	33.80
Total Saleable Production [3]	MMBOE	32.92	32.60	34.99	33.97	31.18

[1] Total production includes crude oil, condensates, NGL, LPG, and total produced gas used for reinjection, artificial lifting, fuel gas, and flares.

[2] Total gas production without gas used for reinjection and artificial lifts.

[3] Saleable Production includes fuel gas exported to PK0, crude oil, condensates, NGL, and LPG.



HBNS GHG EMISSIONS BY MODULE – 2021 TO 2025

Total CO₂e Emissions (Tonnes/Year)



SANGEA Module	2021	2022	2023	2024	2025
Stationary Combustion	692,694.21	712,520.69	715,852.19	552,528.86	540,740.16
Flares*	284,806.75	244,051.90	176,154.42	202,653.29	138,247.85
Equipment Leaks**	35,072.07	33,940.36	33,890.67	1,056.95	2,494.63
Oil and Gas Venting***	12,502.08	13,485.24	13,484.33	15,597.37	15,035.33
Mobile & Transportation	1,512.71	1,949.13	5,839.58	5,838.38	7,293.94
Storage Tank	1,323.22	1,582.79	1,627.10	1,931.19	2,112.50
Miscellaneous****	52.61	19.05	27.40	25.07	19.26
Indirect (Scope 2)*****	87,220.34	86,435.65	91,303.88	75,803.51	78,086.17
Scope 1 Emissions	1,027,963.65	1,007,549.16	946,875.71	779,631.13	705,943.66
Scope 2 Emissions	87,220.34	86,435.65	91,303.88	75,803.51	78,086.17
Total of GHG Emissions	1,115,183.99	1,093,984.81	1,038,179.59	855,434.64	784,029.83

*In 2025, measured destruction removal efficiency (DRE) results were applied to all CPF flares. This replaces prior years' use of the standard 98% default, with DRE now determined through annual field measurements using VISR camera.

**Transition to measurement based methodology for fugitives leaks in 2025 (versus component type emission factors - 2024) resulted an increase from 2024. By comparison, this highest-accuracy method still represents a reduction relative to the 2021–2023 baseline.

***Oil and Gas Venting are calculated based on the annual total, derived from updated monthly average.

****Miscellaneous Sources: fire suppressant, wastewater treatment.

*****Scope 2 Emissions calculations are based on MWh data provided, along with the PK0 turbine efficiency factor.



ELM GHG EMISSIONS BY MODULE – 2021 TO 2025

Total CO₂e Emissions (Tonnes/Year)



SANGEA Module	2021	2022	2023	2024	2025
Stationary Combustion	557,093.77	585,340.12	720,643.74	695,899.51	694,054.62
Flares*	280,697.29	341,868.29	195,795.53	166,779.72	167,767.37
Equipment Leaks**	68,237.89	70,327.44	71,256.55	1,704.77	5,477.63
Oil and Gas Venting***	10,294.91	11,119.57	11,916.70	13,686.82	13,329.14
Mobile & Transportation	1,302.89	2,356.07	3,306.52	2,752.18	3,347.15
Storage Tank	1,386.77	285.47	1,489.59	1,892.83	1,639.25
Miscellaneous****	17.15	10.89	10.10	9.40	13.69
Indirect (Scope 2)*****	171,947.77	214,554.13	244,599.51	250,473.10	239,226.36
Scope 1 Emissions	919,030.66	1,011,307.84	1,004,418.73	882,725.23	885,628.85
Scope 2 Emissions	171,947.77	214,554.13	244,599.51	250,473.10	239,226.36
Total of GHG Emissions	1,090,978.43	1,225,861.97	1,249,018.24	1,133,198.34	1,124,855.22

*In 2025, measured destruction removal efficiency (DRE) results were applied to all CPF flares. This replaces prior years' use of the standard 98% default, with DRE now determined through annual field measurements using VISR camera.

**Transition to measurement based methodology for fugitives leaks in 2025 (versus component type emission factors - 2024) resulted an increase from 2024. By comparison, this highest-accuracy method still represents a reduction relative to the 2021–2023 baseline.

***Oil and Gas Venting are calculated based on the annual total, derived from updated monthly average.

****Miscellaneous Sources: fire suppressant, wastewater treatment.

*****Scope 2 Emissions calculations are based on MWh data provided, along with the PK0 turbine efficiency factor.



GB TOTAL GHG EMISSIONS BY MODULE – 2021 TO 2024

Total CO₂e Emissions (Tonnes/Year)



SANGEA Module	2021	2022	2023	2024	2025
Stationary Combustion	1,249,787.98	1,297,860.81	1,436,495.93	1,248,428.37	1,234,794.79
Flares*	565,504.04	585,920.19	371,949.95	369,433.01	306,015.22
Equipment Leaks**	103,309.96	104,267.80	105,147.22	2,761.72	7,972.26
Oil and Gas Venting***	22,796.99	24,604.81	25,401.03	29,284.19	28,364.47
Mobile & Transportation	2,815.60	4,305.20	9,146.10	8,590.55	10,641.09
Storage Tank	2,709.99	1,868.26	3,116.69	3,824.02	3,751.75
Miscellaneous****	69.76	29.94	37.50	34.47	32.95
Indirect (Scope 2)*****	259,168.11	300,989.78	335,903.39	326,276.62	317,312.54
Scope 1 Emissions	1,946,994.31	2,018,857.00	1,951,294.44	1,662,356.36	1,591,572.51
Scope 2 Emissions	259,168.11	300,989.78	335,903.39	326,276.62	317,312.54
Total of GHG Emissions	2,206,162.42	2,319,846.78	2,287,197.83	1,988,632.97	1,908,885.05

*In 2025, measured destruction removal efficiency (DRE) results were applied to all CPF flares. This replaces prior years' use of the standard 98% default, with DRE now determined through annual field measurements using VISR camera.

**Transition to measurement based methodology for fugitives leaks in 2025 (versus component type emission factors - 2024) resulted an increase from 2024. By comparison, this highest-accuracy method still represents a reduction relative to the 2021–2023 baseline.

***Oil and Gas Venting are calculated based on the annual total, derived from updated monthly average.

****Miscellaneous Sources: fire suppressant, wastewater treatment.

*****Scope 2 Emissions calculations are based on MWh data provided, along with the PK0 turbine efficiency factor.



HBNS CH4 EMISSIONS BY MODULE – 2021 TO 2024

Total CH4 Emissions (Tonnes/Year)



SANGEA Module	2021	2022	2023	2024	2025
Stationary Combustion	38.12	36.88	37.16	33.40	34.30
Flares*	899.16	770.20	573.21	653.33	37.08
Equipment Leaks**	1,401.76	1,356.56	1,354.57	37.75	89.09
Oil and Gas Venting***	499.62	538.92	538.89	556.60	536.54
Mobile & Transportation	0.39	1.27	1.53	1.01	1.15
Storage Tank	52.93	63.31	65.08	68.97	75.45
Miscellaneous****	0.00	0.00	0.00	0.00	0.00
Indirect (Scope 2)*****	9.98	9.89	10.45	8.72	8.99
Scope 1 Emissions	2,891.98	2,767.15	2,570.45	1,351.06	773.62
Scope 2 Emissions	9.98	9.89	10.45	8.72	8.99
Total of GHG Emissions	2,901.97	2,777.04	2,580.90	1,359.78	782.61

*In 2025, measured destruction removal efficiency (DRE) results were applied to all CPF flares. This replaces prior years' use of the standard 98% default, with DRE now determined through annual field measurements using VISR camera.

**Transition to measurement based methodology for fugitives leaks in 2025 (versus component type emission factors - 2024) resulted an increase from 2024. By comparison, this highest-accuracy method still represents a reduction relative to the 2021–2023 baseline.

***Oil and Gas Venting are calculated based on the annual total, derived from updated monthly average.

****Miscellaneous Sources: fire suppressant, wastewater treatment.

*****Scope 2 Emissions calculations are based on MWh data provided, along with the PK0 turbine efficiency factor.



ELM CH4 EMISSIONS BY MODULE – 2021 TO 2025

Total CH4 Emissions (Tonnes/Year)



SANGEA Module	2021	2022	2023	2024	2025
Stationary Combustion	31.18	34.49	42.20	40.61	39.82
Flares*	988.72	1,328.14	781.30	656.52	76.49
Equipment Leaks**	2,724.10	2,807.79	2,844.54	60.88	195.63
Oil and Gas Venting***	403.32	443.89	475.65	487.88	475.13
Mobile & Transportation	0.38	1.28	1.44	0.86	0.94
Storage Tank	55.47	11.42	59.58	67.60	58.54
Miscellaneous****	0.00	0.00	0.00	0.00	0.00
Indirect (Scope 2)*****	19.68	24.56	28.00	28.82	27.53
Scope 1 Emissions	4,203.17	4,627.02	4,204.71	1,314.36	846.56
Scope 2 Emissions	19.68	24.56	28.00	28.82	27.53
Total of GHG Emissions	4,222.85	4,651.57	4,232.70	1,343.18	874.08

*In 2025, measured destruction removal efficiency (DRE) results were applied to all CPF flares. This replaces prior years' use of the standard 98% default, with DRE now determined through annual field measurements using VISR camera.

**Transition to measurement based methodology for fugitives leaks in 2025 (versus component type emission factors - 2024) resulted an increase from 2024. By comparison, this highest-accuracy method still represents a reduction relative to the 2021–2023 baseline.

***Oil and Gas Venting are calculated based on the annual total, derived from updated monthly average.

****Miscellaneous Sources: fire suppressant, wastewater treatment.

*****Scope 2 Emissions calculations are based on MWh data provided, along with the PK0 turbine efficiency factor.



GB TOTAL CH4 EMISSIONS BY MODULE – 2021 TO 2025

Total CH4 Emissions (Tonnes/Year)



SANGEA Module	2021	2022	2023	2024	2025
Stationary Combustion	69.30	71.37	79.36	74.01	74.12
Flares*	1,887.88	2,098.34	1,354.51	1,309.85	113.57
Equipment Leaks**	4,125.86	4,164.35	4,199.11	98.63	284.72
Oil and Gas Venting***	902.94	982.81	1,014.54	1,044.48	1,011.68
Mobile & Transportation	0.77	2.55	2.97	1.87	2.10
Storage Tank	108.40	74.73	124.66	136.57	133.99
Miscellaneous****	0.00	0.00	0.00	0.00	0.00
Indirect (Scope 2)*****	29.66	34.45	38.45	37.54	36.51
Scope 1 Emissions	7,095.15	7,394.17	6,775.16	2,665.41	1,620.18
Scope 2 Emissions	29.66	34.45	38.45	37.54	36.51
Total of GHG Emissions	7,124.82	7,428.61	6,813.60	2,702.96	1,656.69

*In 2025, measured destruction removal efficiency (DRE) results were applied to all CPF flares. This replaces prior years' use of the standard 98% default, with DRE now determined through annual field measurements using VISR camera.

**Transition to measurement based methodology for fugitives leaks in 2025 (versus component type emission factors - 2024) resulted an increase from 2024. By comparison, this highest-accuracy method still represents a reduction relative to the 2021–2023 baseline.

***Oil and Gas Venting are calculated based on the annual total, derived from updated monthly average.

****Miscellaneous Sources: fire suppressant, wastewater treatment.

*****Scope 2 Emissions calculations are based on MWh data provided, along with the PK0 turbine efficiency factor.



2025 YEARLY CPF FLARING BREAKDOWN

Flaring in KNm3



HBNS CPF Flaring				
Year	Routine Flaring	Non-Routine Flaring	Safety Flaring	Total Flaring
2021	25,428.49	62,429.83	1,819.94	89,678.26
2022	26,765.98	34,736.81	1,932.68	63,435.47
2023	26,842.94	21,860.97	1,959.52	50,663.44
2024	26,450.85	35,233.83	1,905.84	63,590.44
2025	26,819.69	18,971.93	1,959.52	47,751.14

ELM CPF Flaring				
Year	Routine Flaring	Non-Routine Flaring	Safety Flaring	Total Flaring
2021	29,081.24	37,938.84	29,365.70	96,385.79
2022	43,816.61	72,950.32	5,256.74	122,023.67
2023	46,806.84	18,628.80	4,225.82	69,661.46
2024	46,814.62	8,779.38	4,237.40	59,831.40
2025	44,786.50	16,369.70	4,225.82	65,382.03

*From 2022 onwards, the routine, non-routine, and safety flaring classification for ELM was modified and updated using recorded events and volumes.



HBNS

2021 TO 2025

TOTAL

PRODUCTION, GHG,

METHANE, FLARE

VOLUMES,

& INTENSITIES

[1] Flare volume based on data provided for GHG emissions. Flaring volume includes both CPF and field flaring.

[2] 2025 data on Routine , Non-routine and Safety flaring volumes are associated with CPF activities; classifications does not include field flaring.

[3] Total production includes oil, NGL, LPG and gas used for reinjection, artificial lifts, fuel gas, & flares.

[4] Saleable Production includes fuel gas export to PK0, crude oil, NGL, and LPG.

[5] Methane intensity based on the Natural Gas Sustainability Initiative (NGSI) Methane Intensity Protocol.

Category	Unit	2021	2022	2023	2024	2025
GHG Emissions						
GHG Direct Emissions (Scope 1)	MMtCO ₂ e	1.028	1.008	0.947	0.780	0.706
GHG Indirect Emissions (Scope 2)	MMtCO ₂ e	0.087	0.086	0.091	0.076	0.078
GHG Emissions (Total Scope 1+ 2)	MMtCO ₂ e	1.115	1.094	1.038	0.855	0.784
Methane Emissions						
Methane Emissions in CH ₄	tCH ₄	2,901.97	2,777.04	2,580.90	1,359.78	782.61
Methane Emissions in CO ₂ e	tCO ₂ e	72,549.20	69,426.01	64,522.42	33,994.48	21,912.98
Flare Volume & Emissions						
Flare Volume ^[1]	MMSm ³	96.965	83.192	61.099	69.638	51.339
Routine Flaring ^[2]	MMSm ³	50.125	46.300	38.192	27.904	28.292
Non-Routine Flaring ^[2]	MMSm ³	0.344	0.289	0.291	37.170	20.014
Safety Flaring ^[2]	MMSm ³	46.496	36.602	22.616	2.011	2.067
Flare GHG Emissions	MMtCO ₂ e	0.285	0.244	0.176	0.203	0.138
Production & Intensity by Total Production^[3]						
Total Production	MMBOE	67.111	72.183	71.148	69.853	70.160
Scope 1 GHG Intensity	tCO ₂ e/BOE	0.0153	0.0140	0.0133	0.0112	0.0101
Scope 2 GHG Intensity	tCO ₂ e/BOE	0.0013	0.0012	0.0013	0.0011	0.0011
Total GHG Intensity	tCO ₂ e/BOE	0.0166	0.0152	0.0146	0.0122	0.0112
Methane Intensity	tCH ₄ /BOE	0.000043	0.000038	0.000036	0.000019	0.000011
Flaring Intensity (Total Flare/Total Gas Produced)	Vol.%	1.91%	1.51%	1.10%	1.22%	0.93%
Production & Intensity by Total Saleable Production^[4]						
Total Saleable Production	MMBOE	28.360	30.453	28.773	24.497	25.435
Scope 1 GHG Intensity	tCO ₂ e/BOE	0.0362	0.0331	0.0329	0.0318	0.0278
Scope 2 GHG Intensity	tCO ₂ e/BOE	0.0031	0.0028	0.0032	0.00309	0.0031
Total GHG Intensity	tCO ₂ e/BOE	0.0393	0.0359	0.0361	0.0349	0.0308
Methane Intensity	tCH ₄ /BOE	0.000102	0.000091	0.000090	0.000056	0.000031
Flaring Intensity	Sm ³ /BOE	3.419	2.732	2.123	2.843	2.018
Methane Intensity per NGSI ^[5]	wt%	0.166%	0.149%	0.147%	0.092%	0.051%



ELM

2021 TO 2025

TOTAL

PRODUCTION, GHG,

METHANE, FLARE

VOLUMES,

& INTENSITIES

Category	Unit	2021	2022	2023	2024	2025
GHG Emissions						
GHG Direct Emissions (Scope 1)	MMtCO2e	0.919	1.011	1.004	0.883	0.886
GHG Indirect Emissions (Scope 2)	MMtCO2e	0.172	0.215	0.245	0.250	0.239
GHG Emissions (Total Scope 1+ 2)	MMtCO2e	1.091	1.226	1.249	1.133	1.125
Methane Emissions						
Methane Emissions in CH4	tCH4	4,222.85	4,651.57	4,232.70	1,343.18	874.08
Methane Emissions in CO2e	tCO2e	105,571.27	116,289.37	105,817.61	33,579.48	24,474.34
Flare Volume & Emissions						
Flare Volume ^[1]	MMSm3	101.682	129.575	75.558	63.344	70.148
Routine Flaring ^[2]	MMSm3	50.271	37.066	43.579	49.387	47.246
Non-Routine Flaring ^[2]	MMSm3	3.026	2.393	3.134	9.262	17.269
Safety Flaring ^[2]	MMSm3	48.384	90.116	28.845	4.470	4.458
Flare GHG Emissions	MMtCO2e	0.281	0.342	0.196	0.167	0.168
Production & Intensity by Total Production^[3]						
Total Production	MMBOE	59.192	60.709	64.742	64.429	60.767
Scope 1 GHG Intensity	tCO2e/BOE	0.0155	0.0167	0.0155	0.0137	0.0146
Scope 2 GHG Intensity	tCO2e/BOE	0.0029	0.0035	0.0038	0.0039	0.0039
Total GHG Intensity	tCO2e/BOE	0.0184	0.0202	0.0193	0.0176	0.0185
Methane Intensity	tCH4/BOE	0.000071	0.000077	0.000065	0.000021	0.000014
Flaring Intensity (Total Flare/Total Gas Produced)	Vol.%	2.78%	3.27%	1.78%	1.45%	1.65%
Production & Intensity by Total Saleable Production^[4]						
Total Saleable Production	MMBOE	32.922	32.605	34.993	33.971	31.184
Scope 1 GHG Intensity	tCO2e/BOE	0.0279	0.0310	0.0287	0.0260	0.0284
Scope 2 GHG Intensity	tCO2e/BOE	0.0052	0.0066	0.0070	0.00737	0.0077
Total GHG Intensity	tCO2e/BOE	0.0331	0.0376	0.0357	0.0334	0.0361
Methane Intensity	tCH4/BOE	0.000128	0.000143	0.000121	0.000040	0.000028
Flaring Intensity	Sm3/BOE	3.089	3.974	2.159	1.865	2.250
Methane Intensity per NGSI ^[5]	wt%	0.164%	0.176%	0.148%	0.047%	0.033%

[1] Flare volume based on data provided for GHG emissions. Flaring volume includes both CPF and field flaring.

[2] 2025 data on Routine , Non-routine and Safety flaring volumes are associated with CPF activities; classifications does not include field flaring.

[3] Total production includes oil, NGL, LPG and gas used for reinjection, artificial lifts, fuel gas, & flares.

[4] Saleable Production includes fuel gas export to PK0, crude oil, NGL, and LPG.

[5] Methane intensity based on the Natural Gas Sustainability Initiative (NGSI) Methane Intensity Protocol.



GB

2021 TO 2025

TOTAL

PRODUCTION, GHG,

METHANE, FLARE

VOLUMES,

& INTENSITIES

[1] Flare volume based on data provided for GHG emissions. Flaring volume includes both CPF and field flaring.

[2] 2025 data on Routine , Non-routine and Safety flaring volumes are associated with CPF activities; classifications does not include field flaring.

[3] Total production includes oil, NGL, LPG and gas used for reinjection, artificial lifts, fuel gas, & flares.

[4] Saleable Production includes fuel gas export to PK0, crude oil, NGL, and LPG.

[5] Methane intensity based on the Natural Gas Sustainability Initiative (NGSI) Methane Intensity Protocol.

Category	Unit	2021	2022	2023	2024	2025
GHG Emissions						
GHG Direct Emissions (Scope 1)	MMtCO2e	1.947	2.019	1.951	1.662	1.592
GHG Indirect Emissions (Scope 2)	MMtCO2e	0.259	0.301	0.336	0.326	0.317
GHG Emissions (Total Scope 1+ 2)	MMtCO2e	2.206	2.320	2.287	1.989	1.909
Methane Emissions						
Methane Emissions in CH4	tCH4	7124.82	7428.62	6,813.60	2,702.96	1656.69
Methane Emissions in CO2e	tCO2e	178,120.46	185715.38	170,340.04	67,573.96	46,387.32
Flare Volume & Emissions						
Flare Volume ^[1]	MMSm3	196.296	212.767	136.657	132.982	121.488
Routine Flaring ^[2]	MMSm3	100.397	83.367	81.771	77.291	75.538
Non-Routine Flaring ^[2]	MMSm3	3.370	2.682	3.425	46.432	37.282
Safety Flaring ^[2]	MMSm3	94.880	126.718	51.461	6.481	6.525
Flare GHG Emissions	MMtCO2e	0.566	0.586	0.372	0.369	0.306
Production & Intensity by Total Production^[3]						
Total Production	MMBOE	126.303	132.893	135.890	134.282	130.927
Scope 1 GHG Intensity	tCO2e/BOE	0.0154	0.0152	0.0144	0.0124	0.0122
Scope 2 GHG Intensity	tCO2e/BOE	0.0021	0.0023	0.0025	0.0024	0.0024
Total GHG Intensity	tCO2e/BOE	0.0175	0.0175	0.0168	0.0148	0.0146
Methane Intensity	tCH4/BOE	0.000056	0.000056	0.000050	0.000020	0.000013
Flaring Intensity (Total Flare/Total Gas Produced)	Vol.%	2.25%	2.25%	1.40%	1.32%	1.25%
Production & Intensity by Total Saleable Production^[4]						
Total Saleable Production	MMBOE	61.282	63.058	63.766	58.468	56.619
Scope 1 GHG Intensity	tCO2e/BOE	0.0318	0.0320	0.0306	0.0284	0.0281
Scope 2 GHG Intensity	tCO2e/BOE	0.0042	0.0048	0.0053	0.0056	0.0056
Total GHG Intensity	tCO2e/BOE	0.0360	0.0368	0.0359	0.0340	0.0337
Methane Intensity	tCH4/BOE	0.000116	0.000118	0.000107	0.000046	0.000029
Flaring Intensity	Sm3/BOE	3.203	3.374	2.143	2.274	2.146
Methane Intensity per NGSI ^[5]	wt%	0.165%	0.158%	0.147%	0.072%	0.041%



TOTAL GHG EMISSIONS BY JV PARTNER EQUITY SHARE

Total CO2e Emissions (Tonnes/Year)

JV Partners	HBNS 2021	ELM 2021	Total 2021	HBNS 2022	ELM 2022	Total 2022	HBNS 2025	ELM 2025	Total 2025
Sonatrach	486,220	510,578	996,798	497,982	556,787	1,054,768	210,120	300,336	510,456
Occidental	233,073	245,470	478,544	239,254	267,483	506,738	244,617	350,955	595,572
Eni	278,796	122,190	400,986	237,176	133,742	370,917	206,984	175,477	382,461
Total Energies	117,094	122,190	239,284	119,573	133,742	253,314	122,309	175,477	297,786
Pertamina	0	58,913	58,913	0	87,159	87,159	0	79,865	79,865
Repsol	0	31,638	31,638	0	46,951	46,951	0	42,744	42,744
Total CO2e	1,115,184	1,090,978	2,206,162	1,093,985	1,225,862	2,319,847	784,030	1,124,855	1,908,885
JV Partners	HBNS 2023*	ELM 2023*	Total 2023*	HBNS 2024	ELM 2024	Total 2024			
Sonatrach	348,148	416,964	765,112	229,256	302,564	531,820			
Occidental	289,066	347,866	636,932	266,896	353,558	620,453			
Eni	256,451	173,933	430,384	225,835	176,779	402,614			
Total Energies	144,514	173,933	318,448	133,448	176,779	310,227			
Pertamina	0	88,725	88,725	0	80,457	80,457			
Repsol	0	47,596	47,596	0	43,062	43,062			
Total CO2e	1,038,180	1,249,018	2,287,198	855,435	1,133,198	1,988,633			

* 2023 Emissions by JV partner calculated based on equity shares changed from 2 May 2023



TOTAL CH4 EMISSIONS BY JV PARTNER EQUITY SHARE

Total CH4 Emissions (Tonnes/Year)

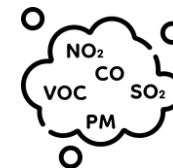
JV Partners	HBNS 2021	ELM 2021	Total 2021	HBNS 2022	ELM 2022	Total 2022	HBNS 2025	ELM 2025	Total 2025
Sonatrach	1,265	1,976	3,242	1,264	2,113	3,377	210	233	443
Occidental	607	950	1,557	607	1,015	1,622	244	273	517
Eni	725	473	1,198	602	507	1,110	207	136	343
Total Energies	305	473	778	304	507	811	122	136	258
Pertamina	0	228	228	0	331	331	0	62	62
Repsol	0	122	122	0	178	178	0	33	33
Total CO2e	2,902	4,223	7,125	2,777	4,652	7,429	783	874	1,657
JV Partners	HBNS 2023*	ELM 2023*	Total 2023*	HBNS 2024	ELM 2024	Total 2024			
Sonatrach	867	1,406	2,273	364	359	723			
Occidental	718	1,182	1,900	424	419	843			
Eni	637	591	1,228	359	210	569			
Total Energies	359	591	950	212	210	422			
Pertamina	0	301	301	0	95	95			
Repsol	0	161	161	0	51	51			
Total CO2e	2,581	4,233	6,814	1,360	1,343	2,703			

* 2023 Emissions by JV partner calculated based on equity shares changed from 2 May 2023



HBNS CAP EMISSIONS BY SOURCE MODULE 2021 TO 2025

CAP Emissions (Tonnes/Year)

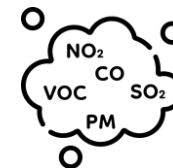


Year	Source Module	NO ₂	CO	SO ₂	PM	VOC
2021	Stationary Combustion	1,574.03	468.64	8.29	36.86	24.47
	Flare	141.17	581.29	0.88	249.12	528.06
	Equipment Leaks	-	-	-	-	364.02
	O&G Venting	-	-	-	-	293.41
	Total	1,715.19	1,049.93	9.17	285.98	1,669.18
2022	Stationary Combustion	1,480.29	467.74	4.09	33.48	20.13
	Flare	121.02	498.32	0.76	213.57	455.40
	Equipment Leaks	-	-	-	-	802.10
	O&G Venting	-	-	-	-	318.65
	Total	1,601.31	966.06	4.84	247.05	1,596.27
2023	Stationary Combustion	1,550.16	484.61	8.67	39.12	26.51
	Flare	87.44	360.03	0.56	154.30	303.44
	Equipment Leaks	-	-	-	-	717.06
	O&G Venting	-	-	-	-	285.27
	Total	1,637.60	844.64	9.22	193.42	1,332.28
2024	Stationary Combustion	1319.25	371.20	5.98	27.27	17.02
	Flare	99.66	410.35	0.63	175.86	345.85
	Equipment Leaks	-	-	-	-	39.10
	O&G Venting	-	-	-	-	294.64
	Total	1418.91	781.55	6.61	203.14	696.62



HBNS CAP EMISSIONS BY SOURCE MODULE 2021 TO 2025

CAP Emissions (Tonnes/Year)

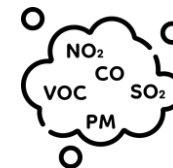


Year	Source Module	NO ₂	CO	SO ₂	PM	VOC
2025	Stationary Combustion	1340.53	363.13	5.87	26.01	15.85
	Flare	74.18	305.44	0.47	130.90	19.63
	Equipment Leaks	-	-	-	-	86.87
	O&G Venting	-	-	-	-	284.03
	Total	1414.70	668.58	6.35	156.91	406.37



ELM CAP EMISSIONS BY SOURCE MODULE 2021 TO 2025

CAP Emissions (Tonnes/Year)

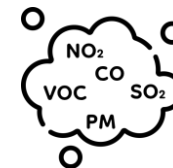


Year	Source Module	NO ₂	CO	SO ₂	PM	VOC
2021	Stationary Combustion	1,306.03	396.94	10.18	34.29	24.63
	Flare	138.7	571.12	0.93	244.76	364.07
	Equipment Leaks	-	-	-	-	373.48
	O&G Venting	-	-	-	-	53.46
	Total	1,444.73	968.06	11.11	279.06	815.64
2022	Stationary Combustion	1424.84	418.04	10.69	35.35	25.02
	Flare	168.75	694.87	1.18	297.8	292.32
	Equipment Leaks	-	-	-	-	307.23
	O&G Venting	-	-	-	-	47.71
	Total	1593.6	1112.9	11.87	333.15	672.28
2023	Stationary Combustion	1727.01	510.26	11.62	41.71	28.69
	Flare	96.74	398.33	0.69	170.71	134.9
	Equipment Leaks	-	-	-	-	491.14
	O&G Venting	-	-	-	-	82.13
	Total	1823.75	908.59	12.31	212.42	736.86
2024	Stationary Combustion	1668.06	494.93	11.6	40.89	28.38
	Flare	81.47	335.47	0.58	143.77	119.47
	Equipment Leaks	-	-	-	-	20.57
	O&G Venting	-	-	-	-	88.78
	Total	1749.53	830.4	12.18	184.66	257.19



ELM CAP EMISSIONS BY SOURCE MODULE 2021 TO 2025

CAP Emissions (Tonnes/Year)



Year	Source Module	NO ₂	CO	SO ₂	PM	VOC
2025	Stationary Combustion	1669.62	500.29	13.43	43.39	31.28
	Flare	91.16	375.36	0.65	160.87	11.11
	Equipment Leaks	-	-	-	-	19.03
	O&G Venting	-	-	-	-	69.03
	Total	1760.78	875.66	14.08	204.26	130.45

